

LAB 9

Wireless Network Setup and Packet Analysis using Cisco Packet Tracer

OBJECTIVES

- To configure a wireless router in Cisco Packet Tracer and create a working wireless network.
- To connect multiple wireless clients to the network and obtain IP addresses automatically using DHCP.
- To apply WPA2 security settings to protect wireless communication.
- To observe and verify packet transmission using Packet Tracer's simulation mode.

THEORY

1. Basics of Wireless Networking

Wireless networks enable communication between devices without the need for physical cables by using radio frequency signals. A wireless router serves as the central node, allowing laptops, smartphones, and other wireless devices to connect. Communication standards defined by IEEE 802.11 ensure compatibility and reliable Wi-Fi connectivity among devices.

2. Functions of a Wireless Router

A wireless router performs multiple roles within a network:

- **Access Point:** Provides wireless connectivity to client devices.
- **Router:** Manages traffic between the local network and external networks.
- **DHCP Server:** Automatically assigns IP addresses to connected devices, simplifying network setup.

In Packet Tracer simulations, routers such as the WRT300N are commonly used to demonstrate wireless networking concepts.

3. Wireless Security Using WPA2

Wireless networks can be vulnerable to unauthorized access if not properly secured. WPA2 is a widely used security protocol that employs AES encryption to protect wireless data. Implementing WPA2 ensures user authentication, maintains data confidentiality, and reduces the risk of unauthorized access or interception.

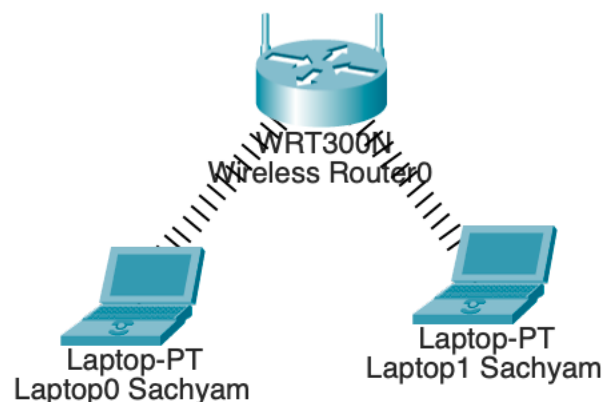
4. Packet Analysis in Packet Tracer

Packet Tracer's simulation mode allows users to visualize how data moves through the network. Through packet analysis, it is possible to observe DHCP exchanges, ARP operations, and encrypted wireless transmissions. This helps reinforce theoretical knowledge with practical verification.

NETWORK TOPOLOGY

The network implemented in this lab follows a simple wireless star topology, where a wireless router acts as the central connection point. All laptops connect to the router using wireless signals, enabling centralized control of IP addressing and security.

This type of topology closely resembles real-world home or small office networks, where multiple devices rely on a single wireless access point for connectivity and network services.



CONFIGURATION

The DHCP assigned IP addresses of the laptops are as follows:

Device	IPv4 Address	Subnet Mask	Default Gateway
Laptop0 Sachyam	192.168.0.101	255.255.255.0	192.168.0.1
Laptop1 Sachyam	192.168.0.102	255.255.255.0	192.168.0.1

Wireless Router Configuration

Step 1: Basic Setup

1. Select the **Wireless Router**.
2. Navigate to **GUI → Setup**.
3. Configure the following parameters:
 - **IP Address:** 192.168.0.1
 - **Subnet Mask:** 255.255.255.0
 - **DHCP Server:** Enabled

Step 2: Wireless Settings

1. Go to **Wireless → Basic Wireless Settings**.
2. Configure:
 - **Network Mode:** Mixed
 - **SSID:** HCOE
 - **Channel:** Auto
3. Click **Save Settings**.

Step 3: Wireless Security

1. Navigate to **Wireless → Wireless Security**.
2. Configure the following:
 - **Security Mode:** WPA2-Personal
 - **Encryption:** AES
 - **Passphrase:** hcoe@123
3. Save the configuration.

Testing

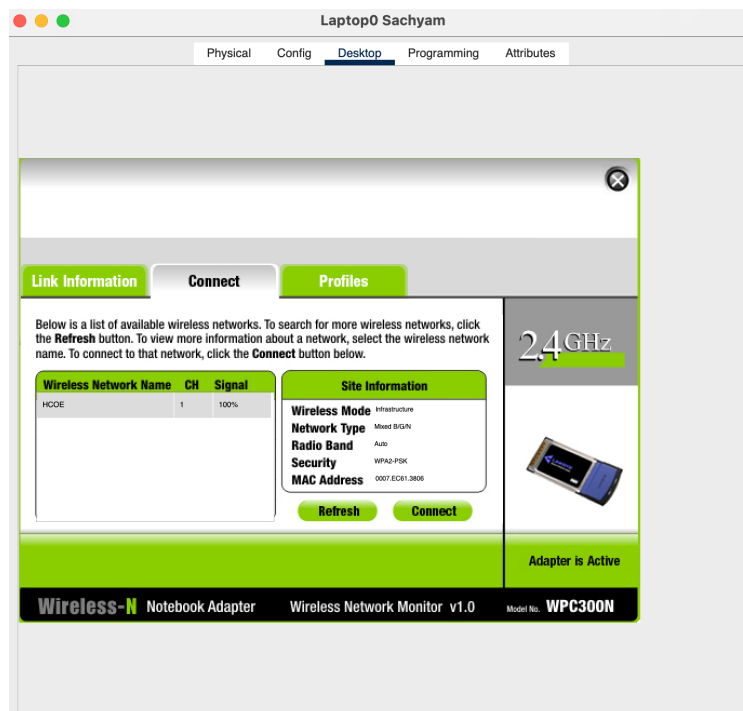


Fig: Connection Established to Network

Ping Test: To test whether both laptops are connected to the wireless network

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Pinging 192.168.0.102 with 32 bytes of data:

Reply from 192.168.0.102: bytes=32 time=28ms TTL=128
Reply from 192.168.0.102: bytes=32 time=22ms TTL=128
Reply from 192.168.0.102: bytes=32 time=20ms TTL=128
Reply from 192.168.0.102: bytes=32 time=20ms TTL=128

Ping statistics for 192.168.0.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 28ms, Average = 22ms
```

DISCUSSION

In this lab, a wireless network was successfully established using a WRT300N router and laptops as client devices in Cisco Packet Tracer. The DHCP service automatically assigned IP addresses to the laptops, minimizing the need for manual configuration and reducing potential errors. WPA2 security was implemented to ensure encrypted and secure wireless communication. Packet analysis in simulation mode verified proper DHCP exchanges and demonstrated how data is transmitted across the wireless network, reinforcing the theoretical concepts with practical observation.

CONCLUSION

This experiment effectively demonstrated the setup and configuration of a wireless network using Cisco Packet Tracer. The use of DHCP enabled dynamic IP address allocation, while WPA2 security ensured protected communication between devices. Packet analysis confirmed the correct functioning of the network, enhancing both conceptual understanding and hands-on skills related to wireless networking technologies.